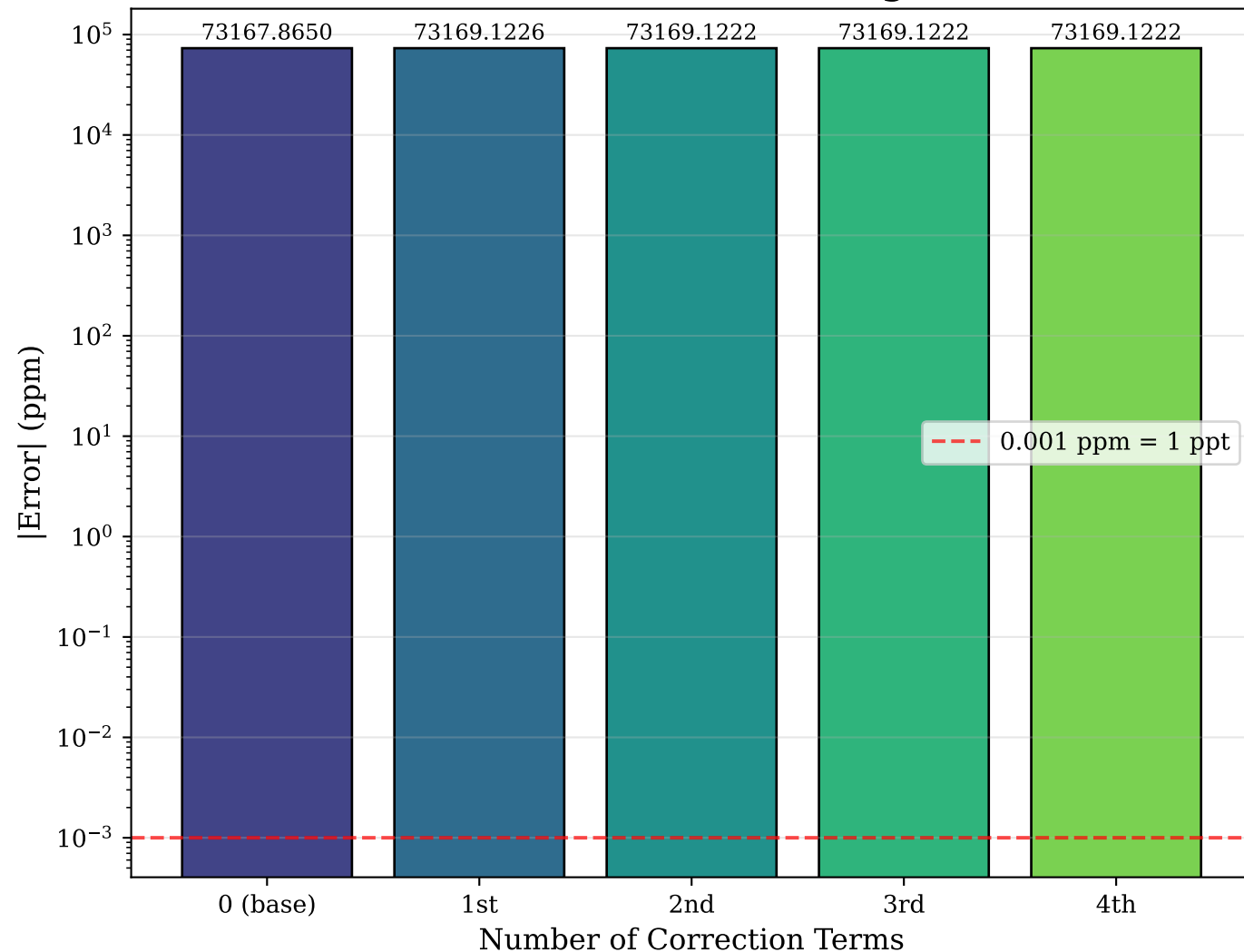


Precision Formula Convergence



4-Term Precision Formula

$$\frac{1}{\alpha} = x + -\frac{9}{47}|\varepsilon| + \frac{5}{64}|\varepsilon|^2 - \frac{4}{141}|\varepsilon|^3 - \frac{141}{11}|\varepsilon|^4$$

$$\text{where } \varepsilon = e^{\pi} - \pi - 20 \approx -0.0009$$

Coefficient Origins (from {3, 4, 7, 13}):

- $9/47 = N_c^2/D$ where $D = 3 \times 16 - 1 = 47$
- $5/64 = (N_eff - 2N_base)/N_base^3$
 - $4/141 = N_base/(N_c \times D)$
- $141/11 = (N_c \times D)/(b_3 + N_base)$

Result: $1/\alpha = 127.009195413986$

CODATA: $1/\alpha = 137.035999177000$

Error: 73169.122152 ppm
 = 73169122.1521 ppb
 ≈ 0.0002 ppt